

VLSI Design & Practice 06

Memory Built-in Self Test (MBIST)

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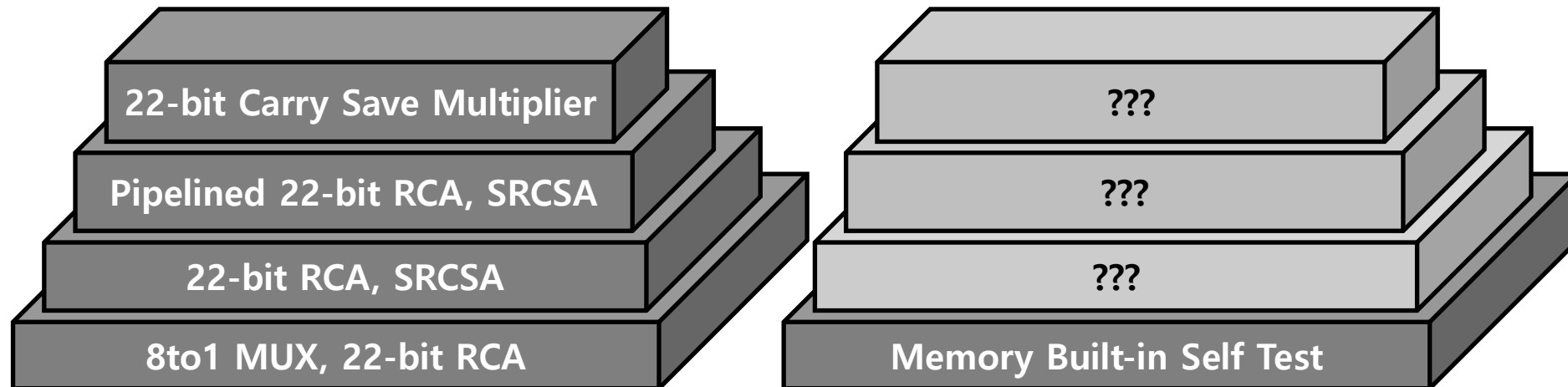
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Contents



**Today we will talk about the
usage of memory !!!**

Contents

1. Design MBIST (Memory Built-in Self Test)

- Introduction of the embedded SRAM Verilog model
- Design MBIST circuit and the MBIST Test Bench
- Check the functionality

2. Memory control with 22×22 multiplier

- Control the memory for 22×22 multiplier operation
- Check the functionality

3. Don't need to Synthesis for today's experiment !!

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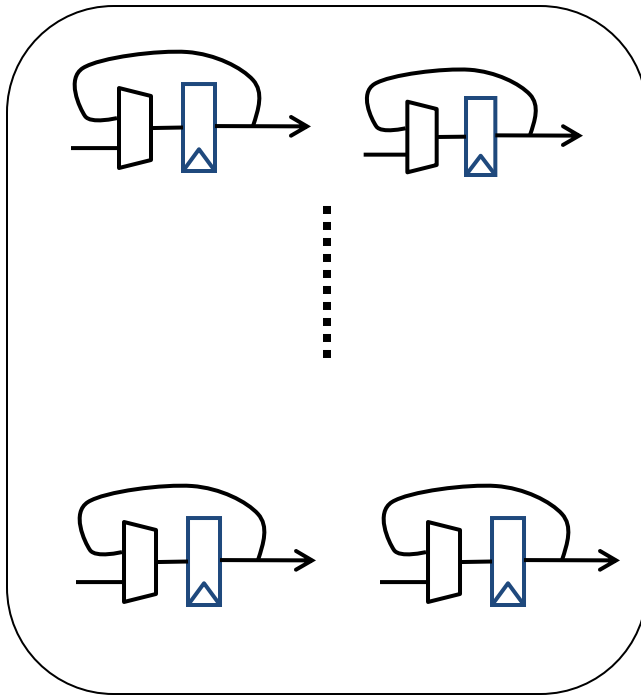
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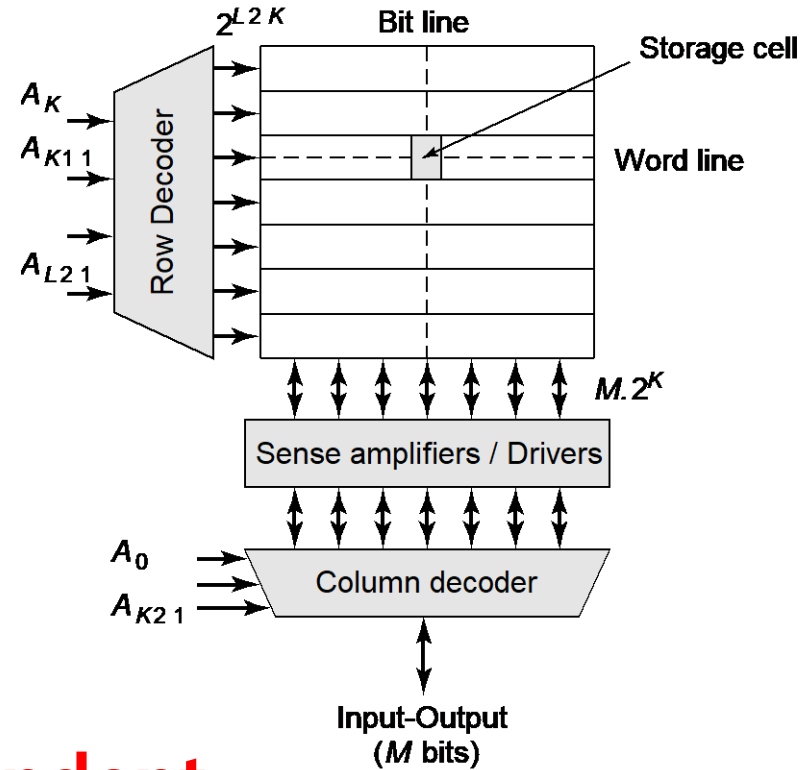
Data Storage

Flip - flops ?



vs.

Memory

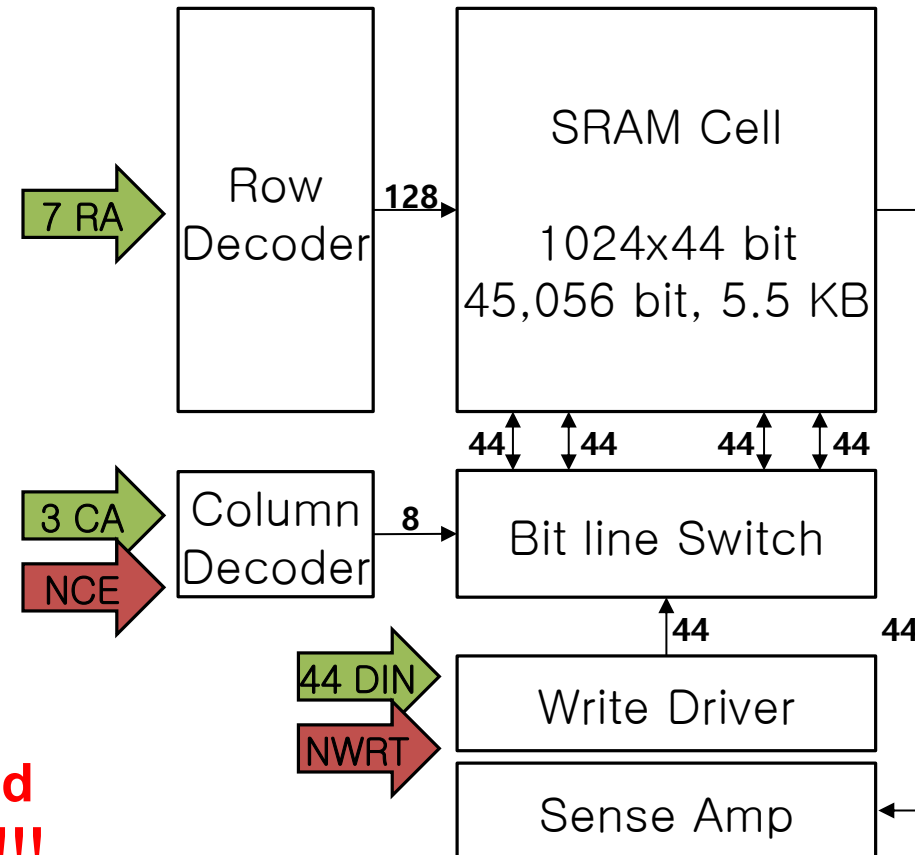


Size dependent

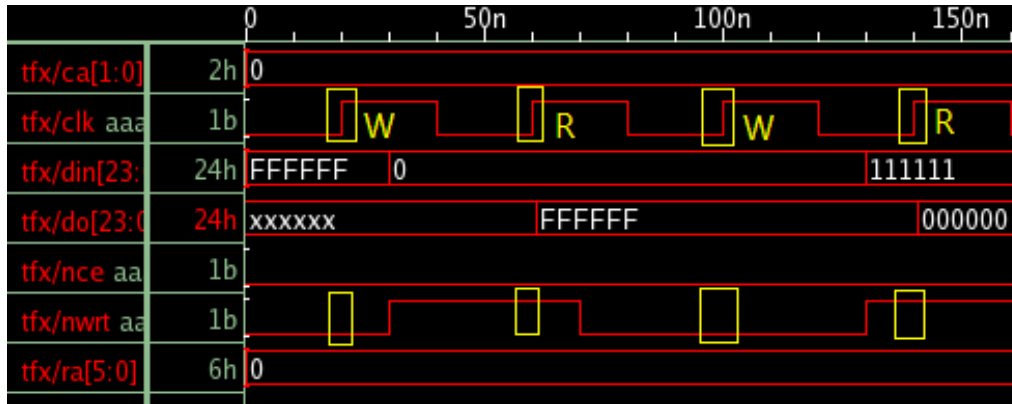
Embedded SRAM Verilog model

- 1 port SRAM(Read or Write)
- Bit Width
 - Input
 - Row Address : 7 bit
 - Column Address : 3 bit
 - Data In : 44 bit
 - Output
 - Data Out : 44bit
 - Control Signal
 - NWRT : 1bit (0 → Write, 1 → Read)
 - NCE : 1bit (0 → Enable, 1 → Disable)
 - Positive edge triggered(CLK)

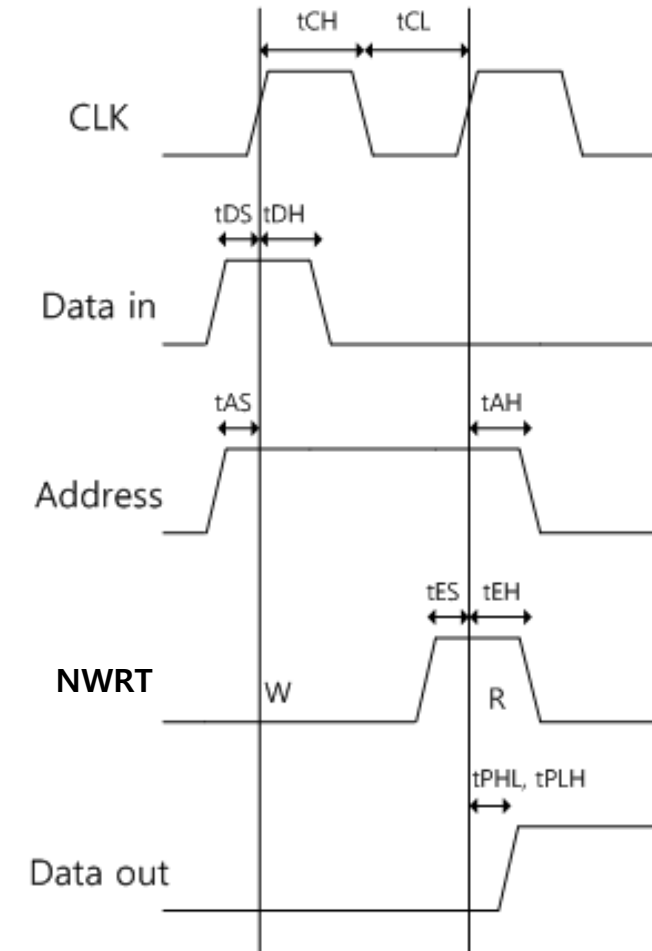
Please use positive edge triggered (CLK) flipflop for this experiment !!!



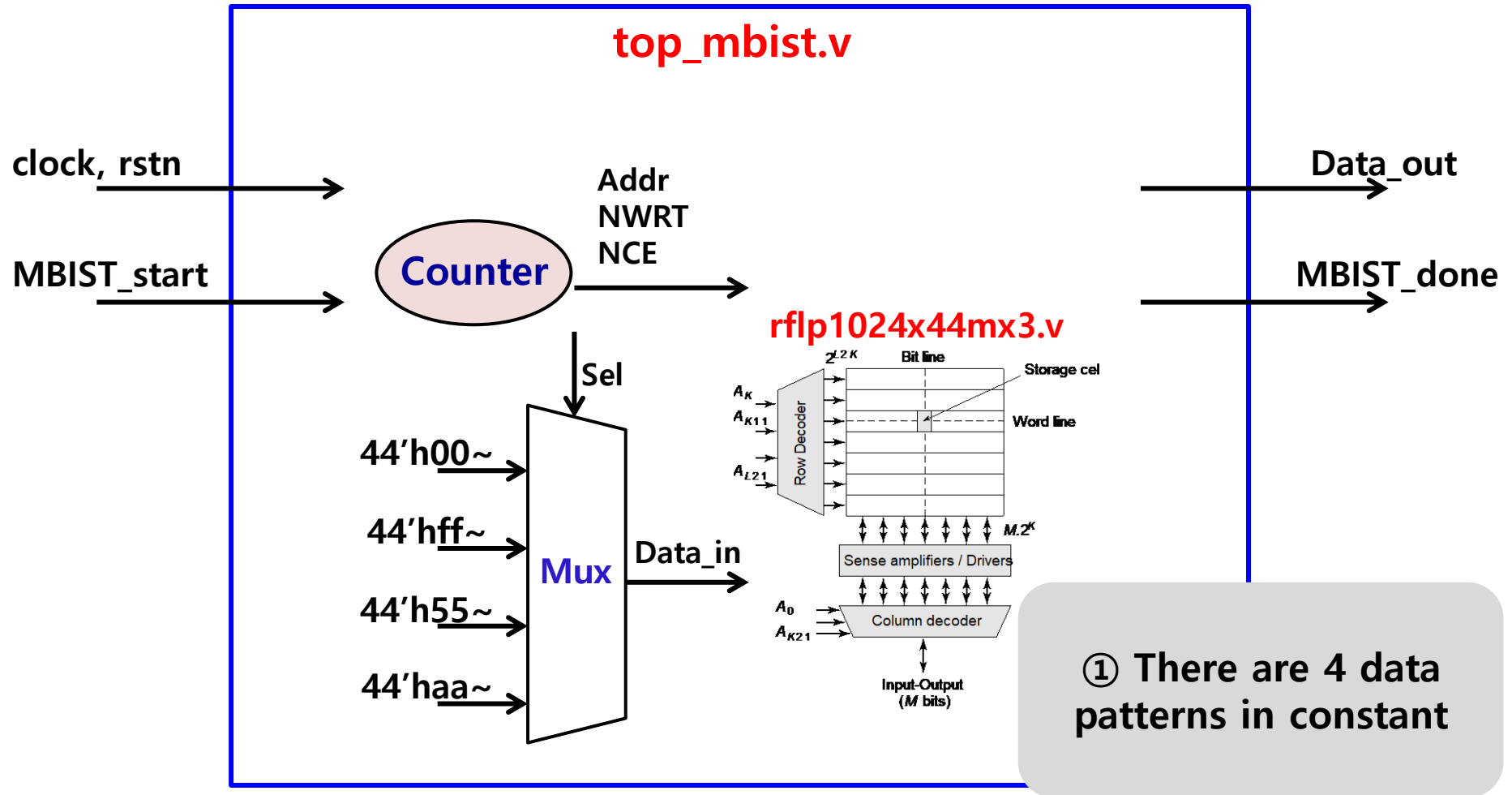
Embedded SRAM Verilog model



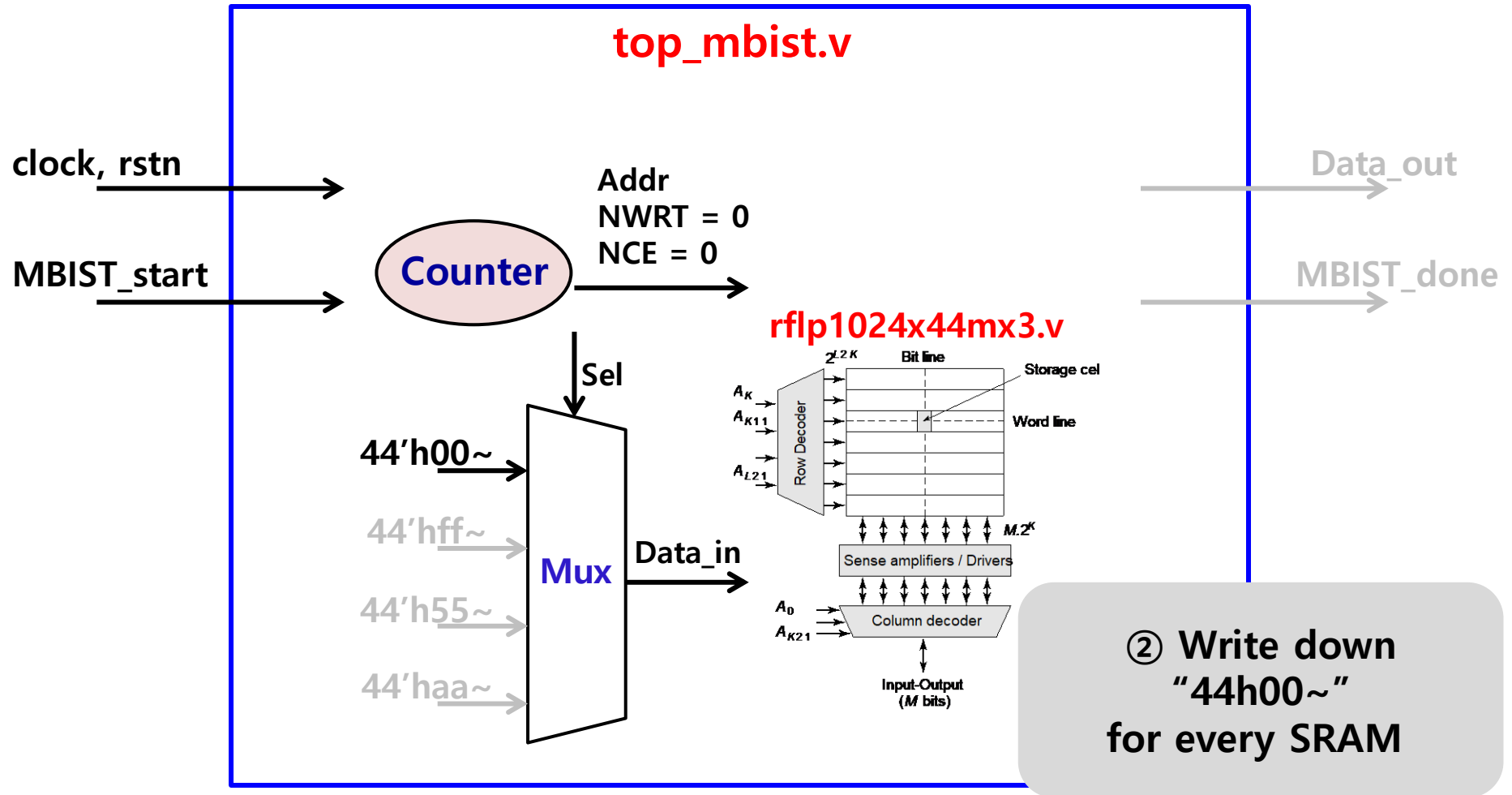
- There are setup time for "Data in" and clk to Q delay for "Data out"
- "Data out" is sustained until the next data read
- NCE(Negative Chip Enable)
- NWRT (Negative WRiTe)



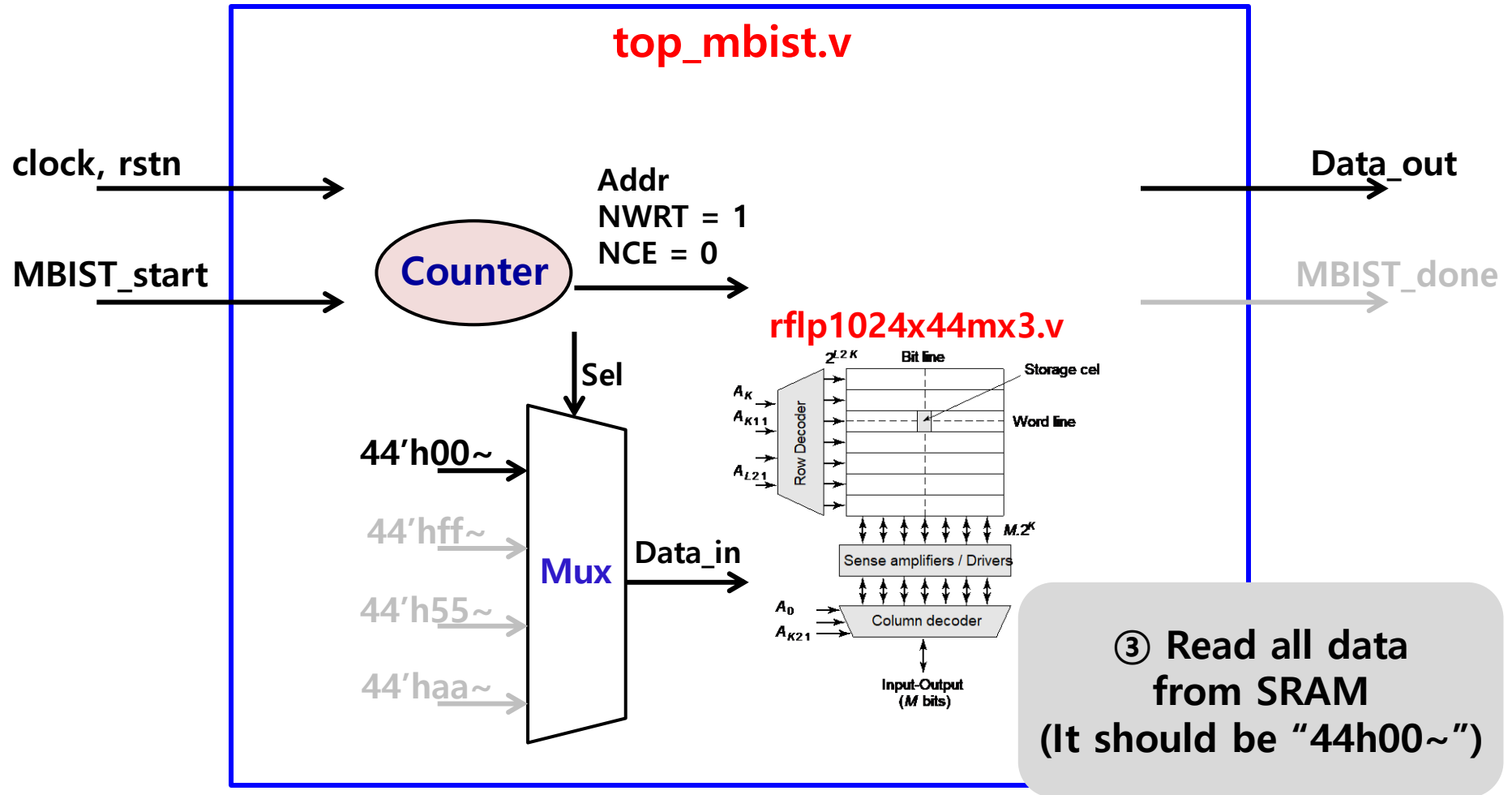
MBIST (Memory Built-in Self Test)



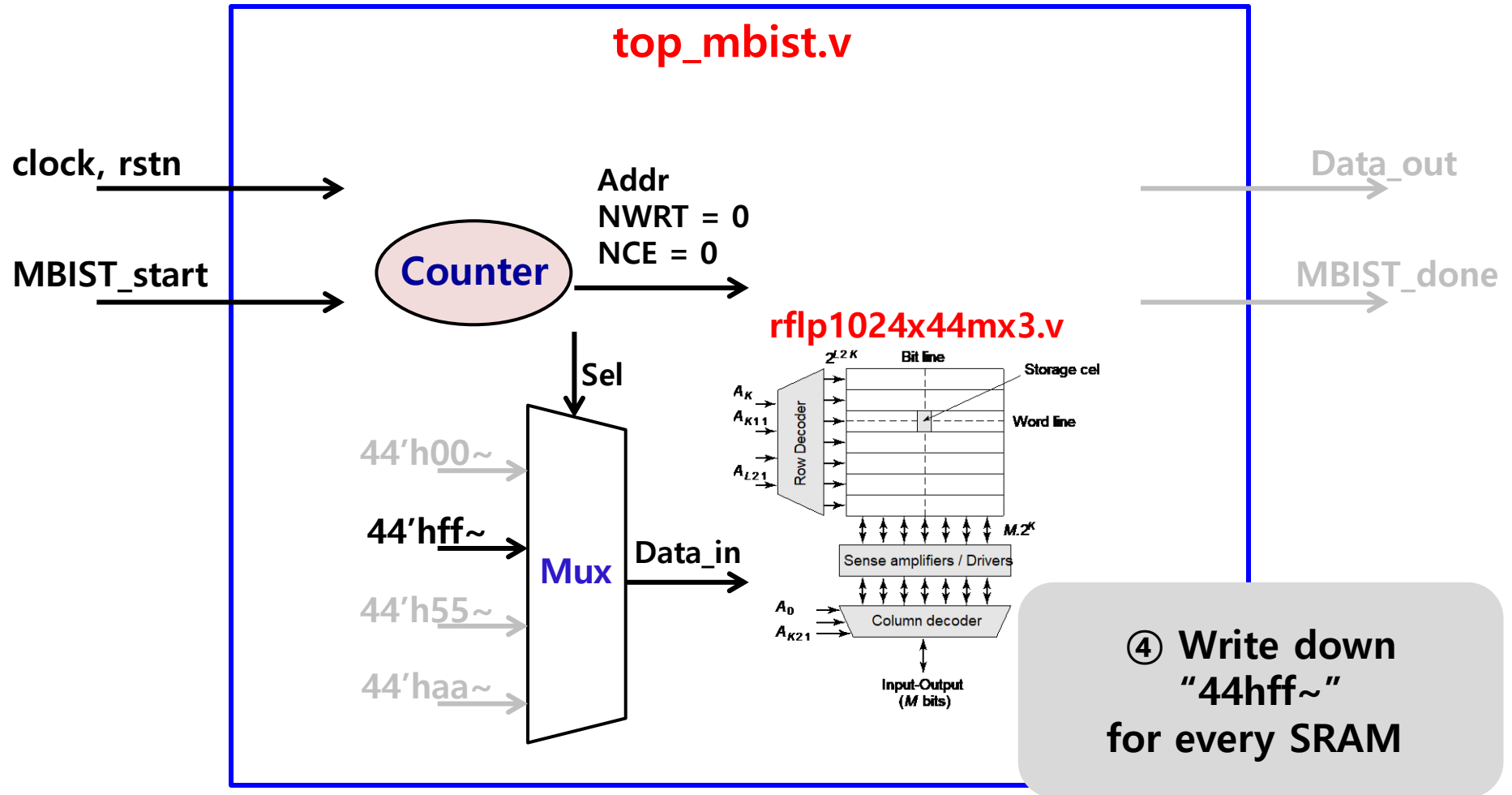
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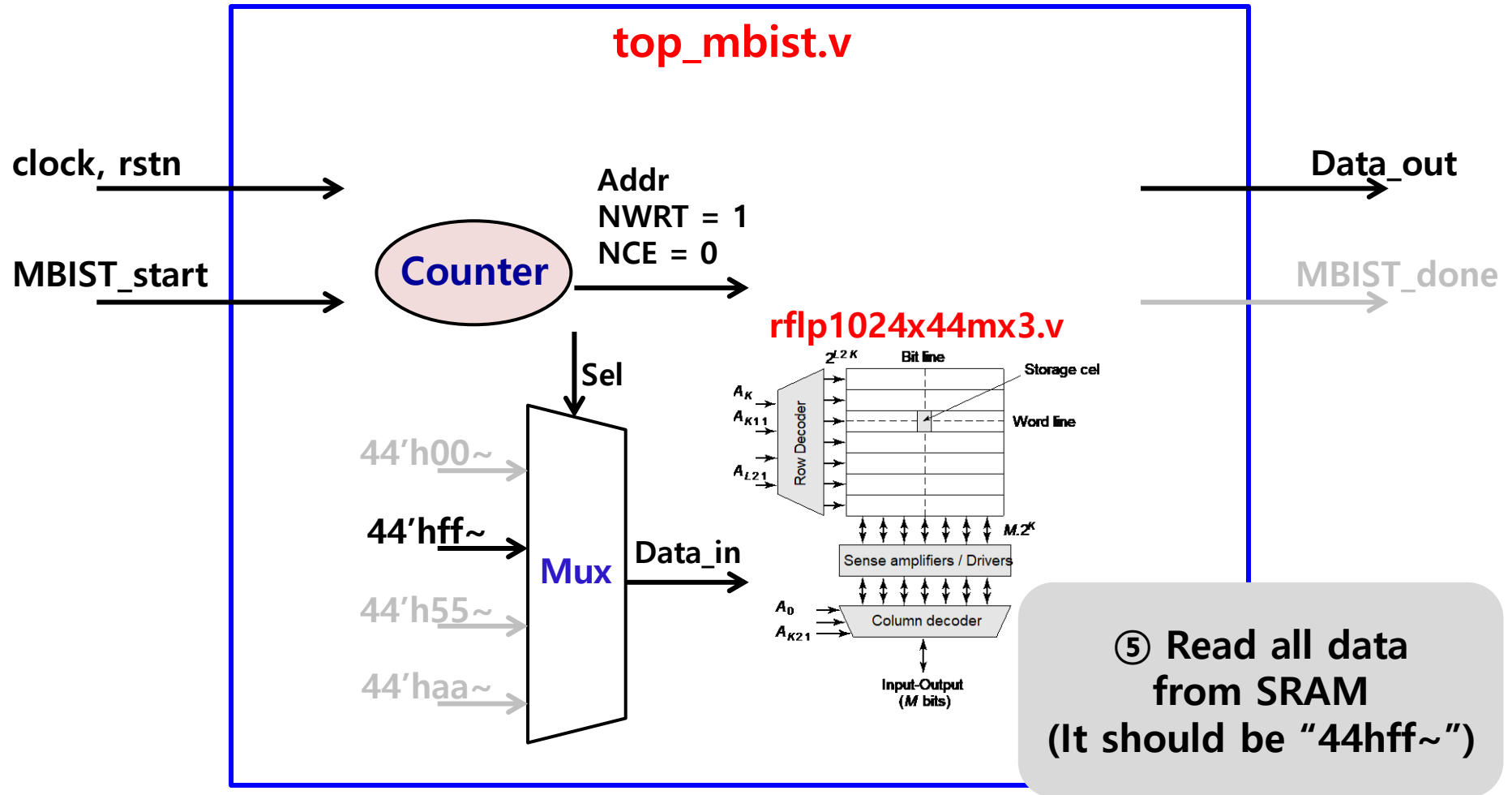
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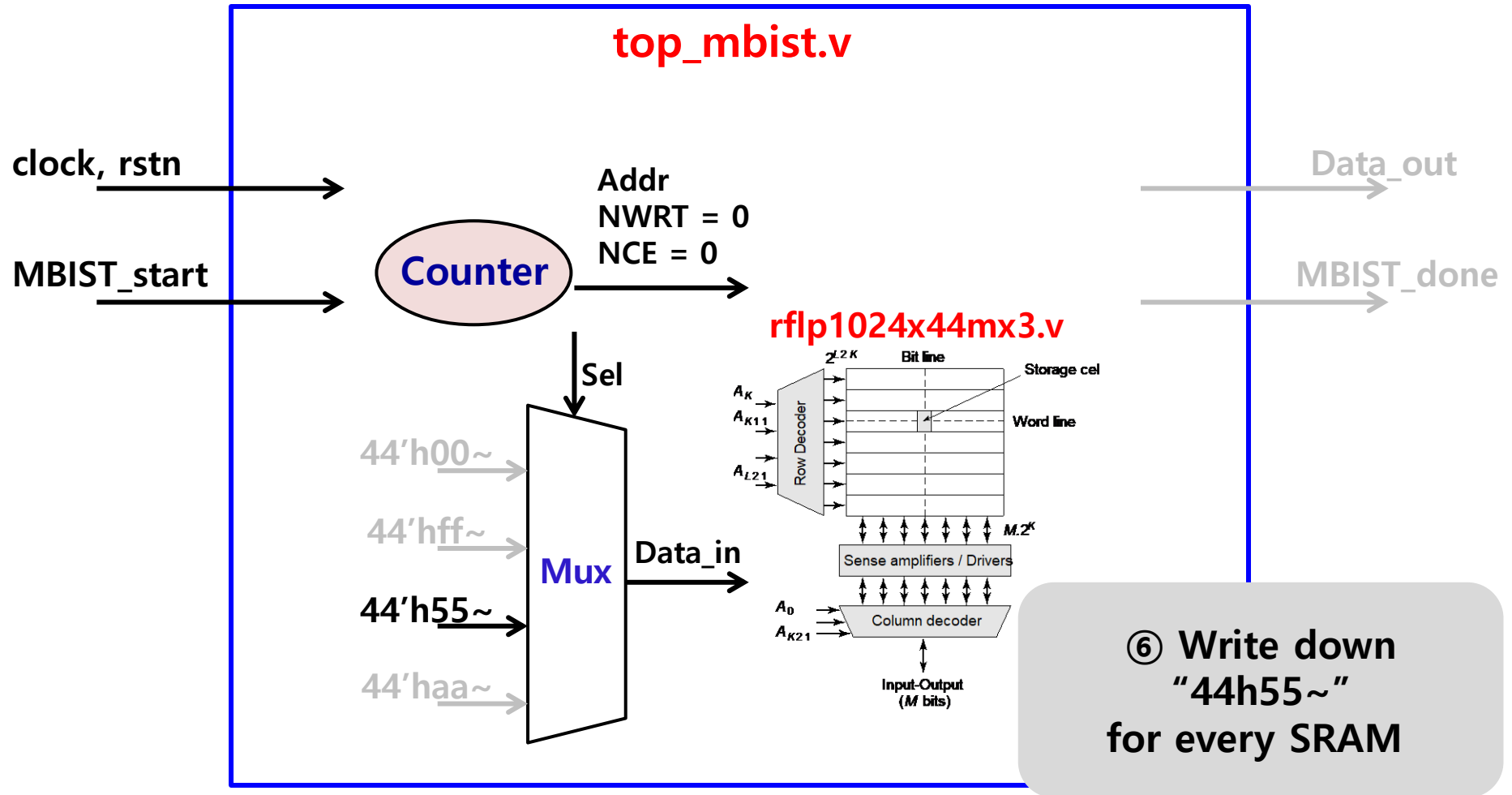
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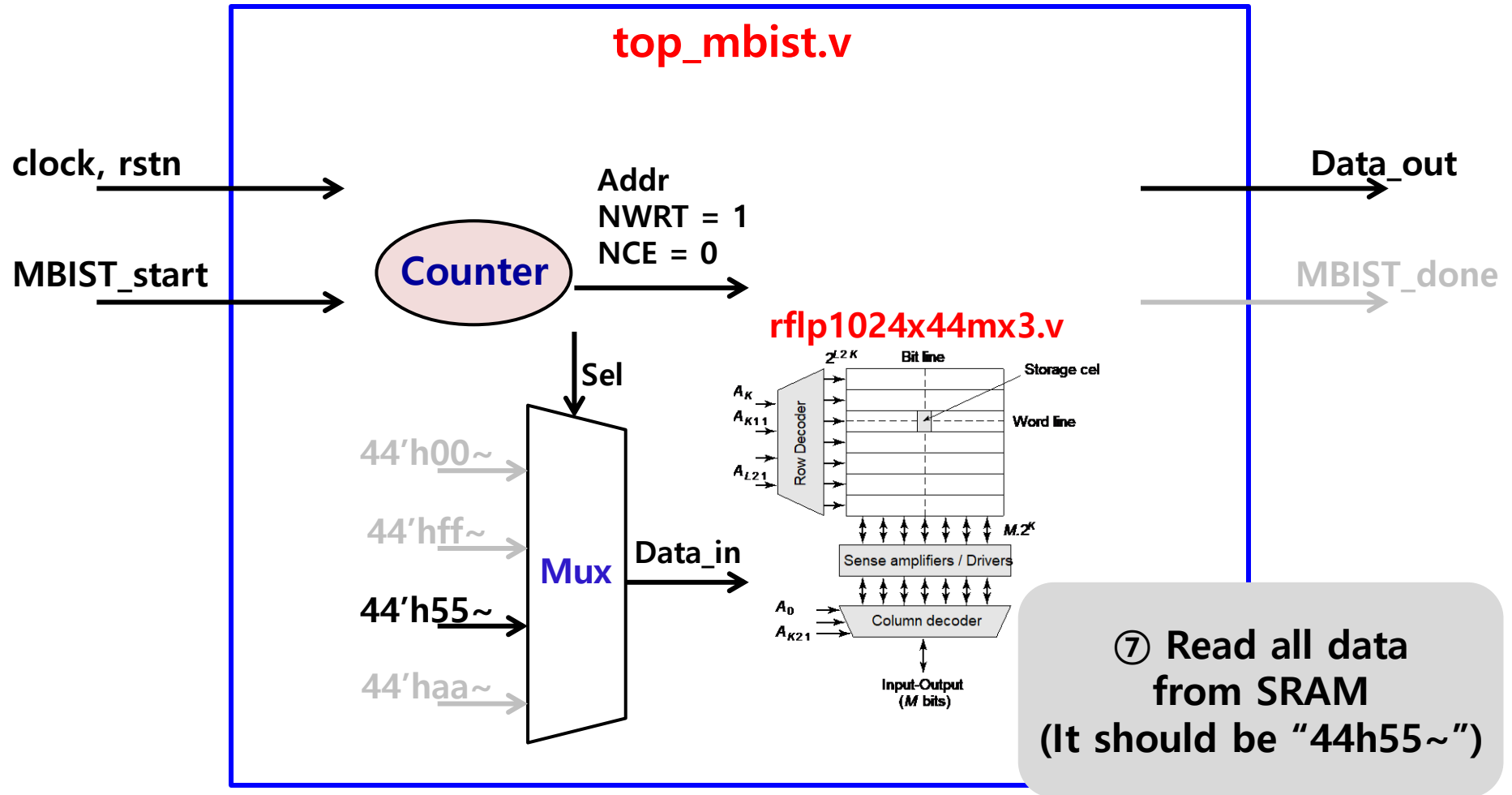
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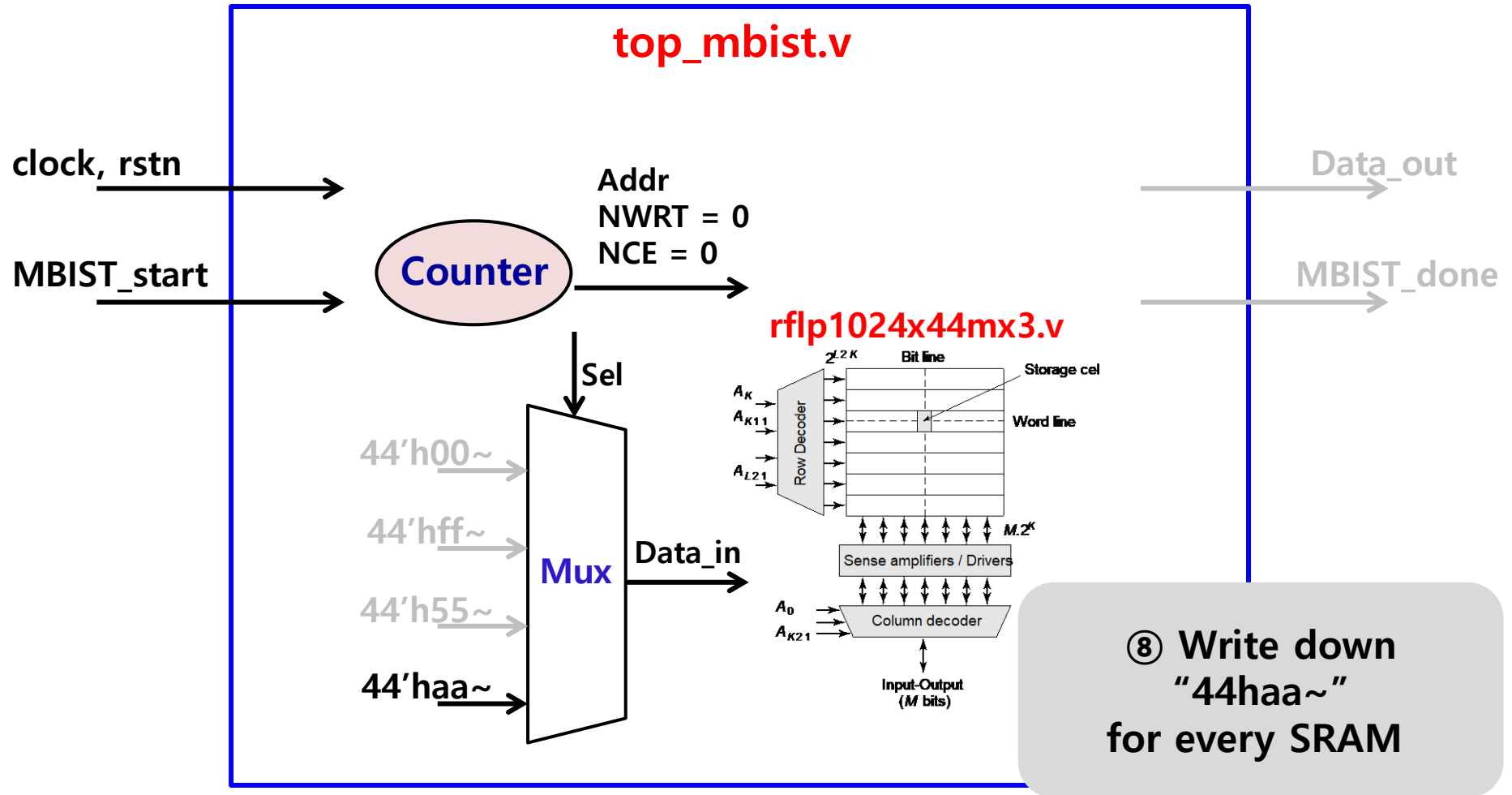
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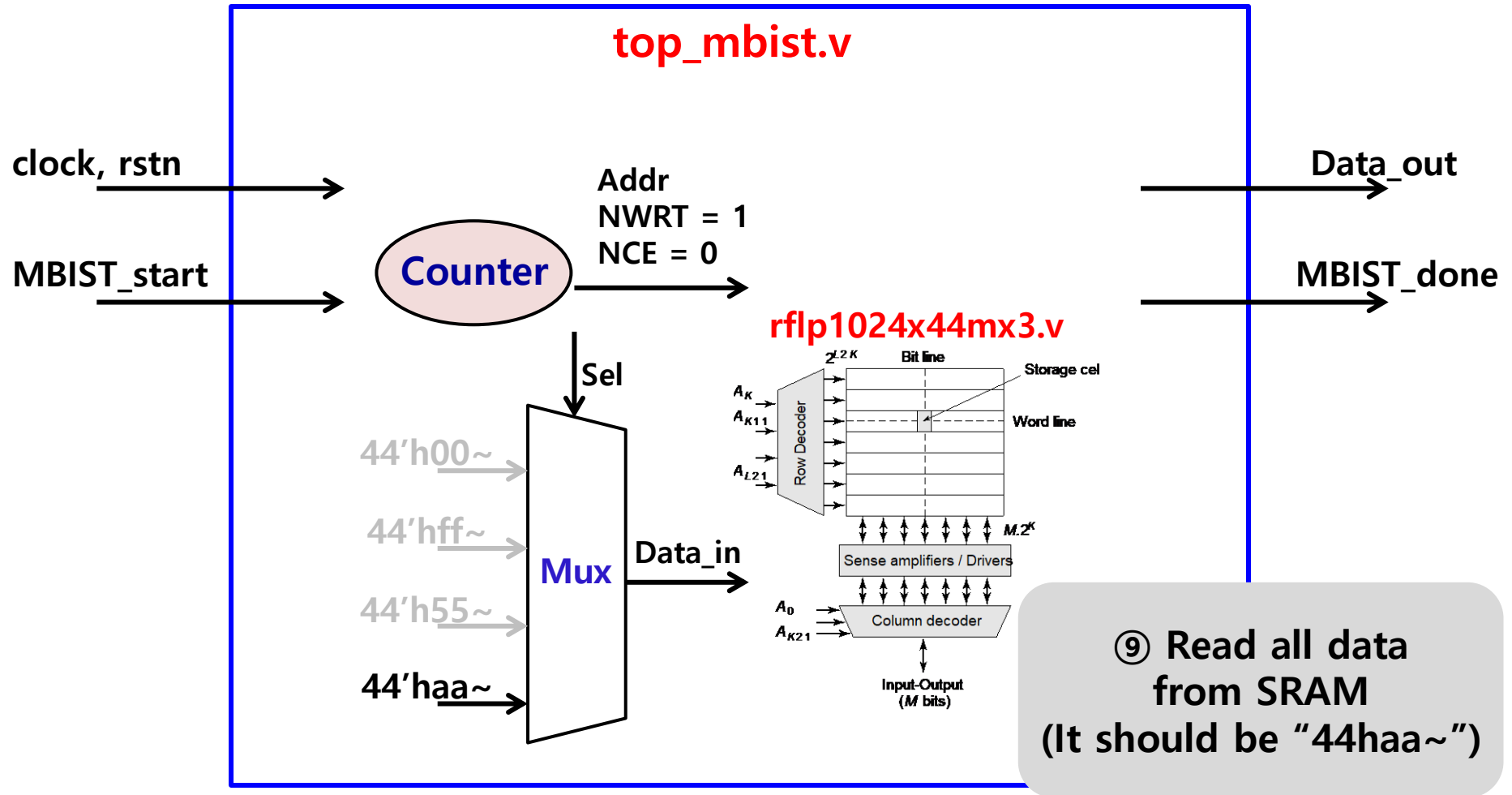
MBIST (Memory Built-in Self Test)



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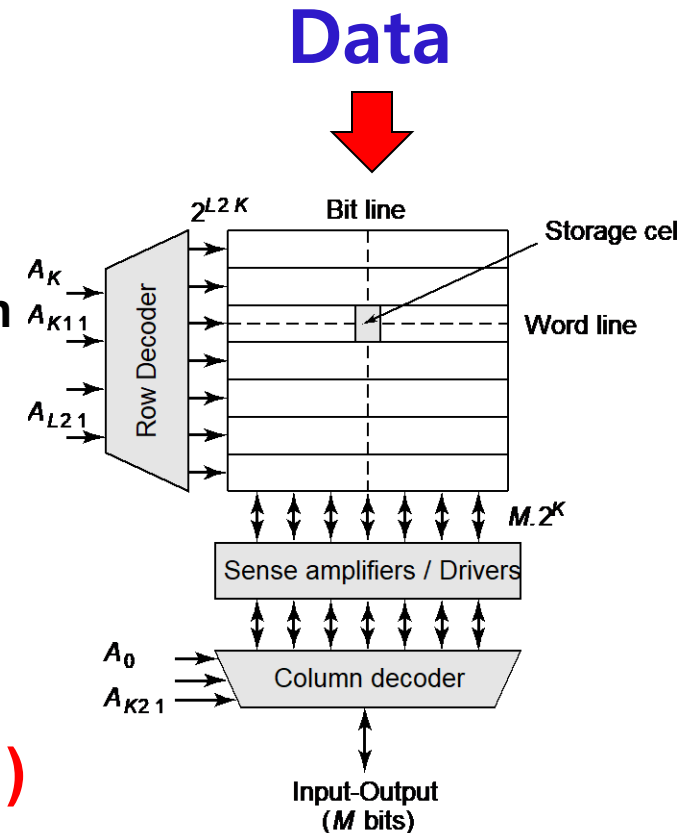
MBIST (Memory Built-in Self Test)

Make the MBIST (Memory Built-in Self Test)
for the memory (rflp1024x44mx3.v) !

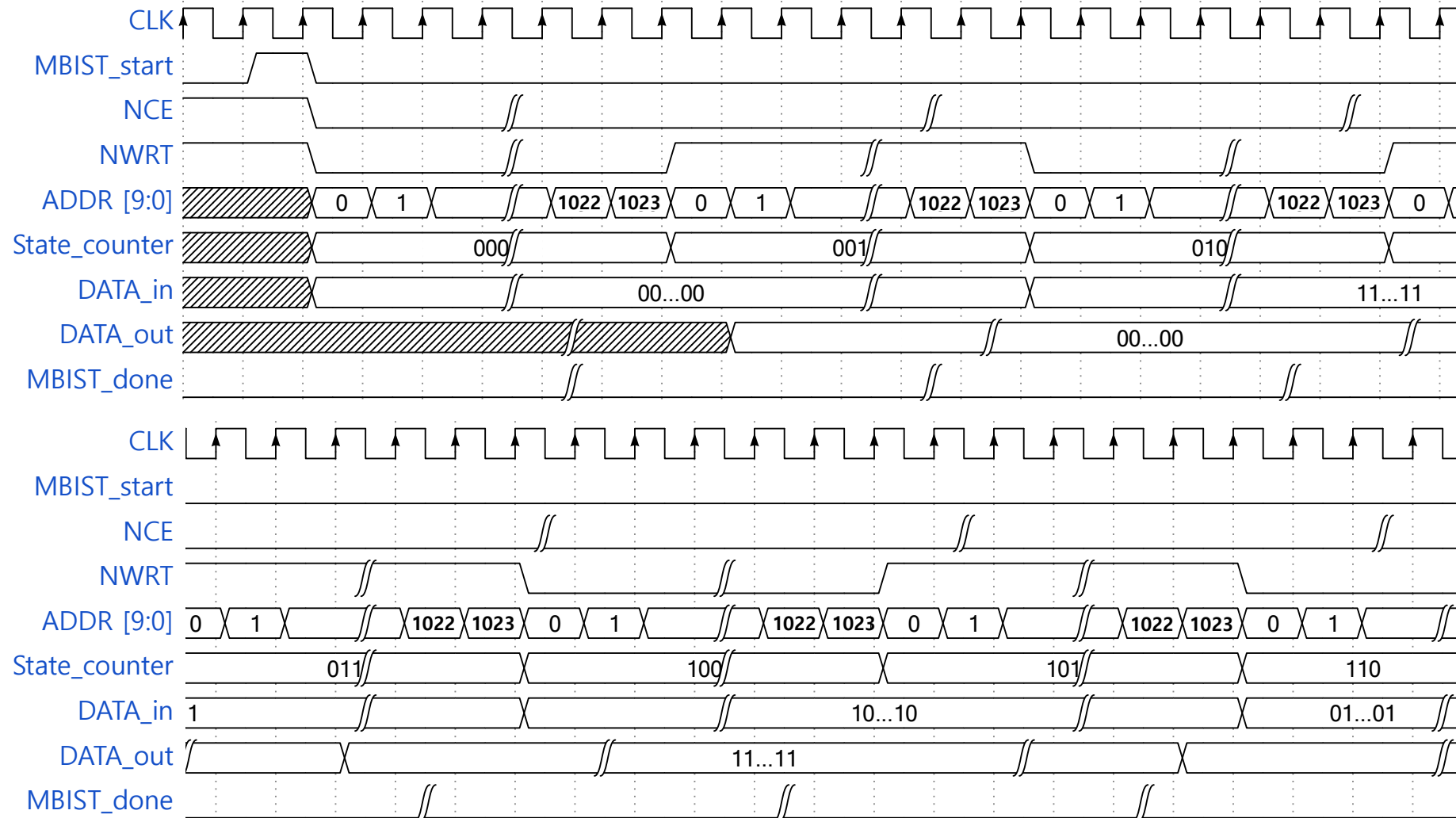
- 1) 1024 x 44 sizes of 0000..., 1111..., 0101..., 1010 ...
- 2) **10-bit counter** is needed for address generation
- 3) **3-bit state counter** is needed for muxing
- 4) Write & Read for 4 patterns

PLEASE MAKE the MBIST Test Bench

**(DO NOT USE FOR STATEMENT
FOR MEMORY CONTROL SIGNAL !!!)**

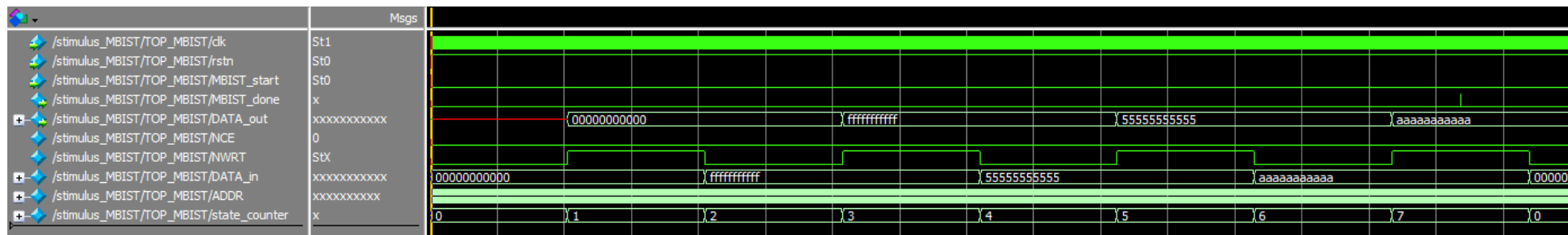


MBIST (Memory Built-in Self Test)



Functional Verifications

- Simulation results



- Check the expected outputs are coming !!!

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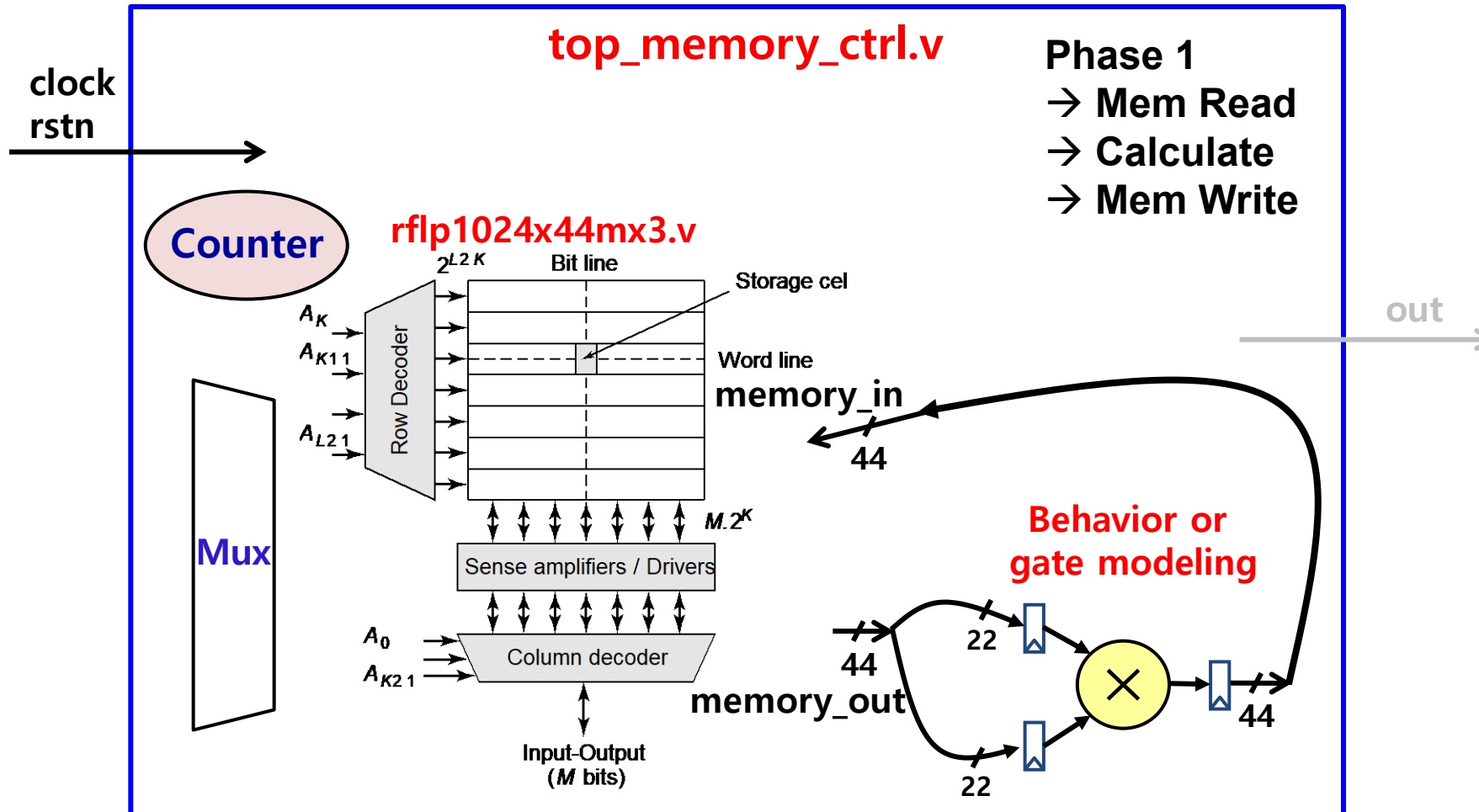
Memory control with 22×22 multiplier

1) Storing input data into SRAM directly

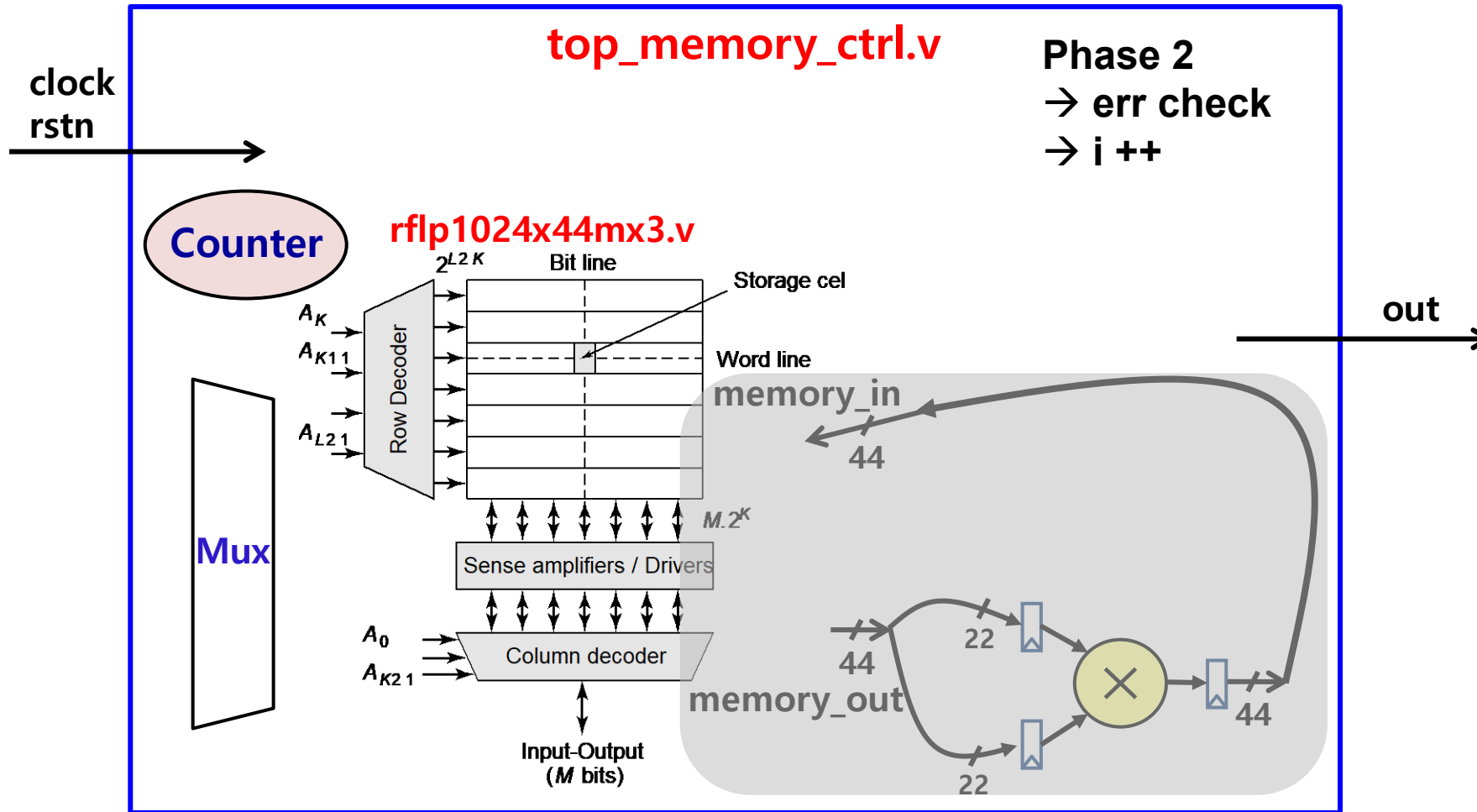
```
initial $readmemh("input.txt", TOP_MEM_TEST.memory.array);  
  
top_memory_ctrl TOP_MEM_TEST(out, clk, rstn);  
  
rflpl024x44mx3 memory.DO(memory_out), .DIN(memory_in), .RA(RA), .CA(CA), .NWRT(nwrt), .NCE(nce), .CLK(clk));
```

- 2) Read 44-bit input data and split it into “A” and “B” Mem Read
- 3) (A → delayed A, B → delayed B) using D filp-flops
- 4) Calculate “mul_out” using “delayed A” and “delayed B” (Behavior modeling ? Gate modeling?)
- 5) (mul_out → delayed mul_out) using D filp-flop Calculate
- 6) Write down “delayed mul_out” into SRAM (same address) Mem Write

Memory control with 22×22 multiplier

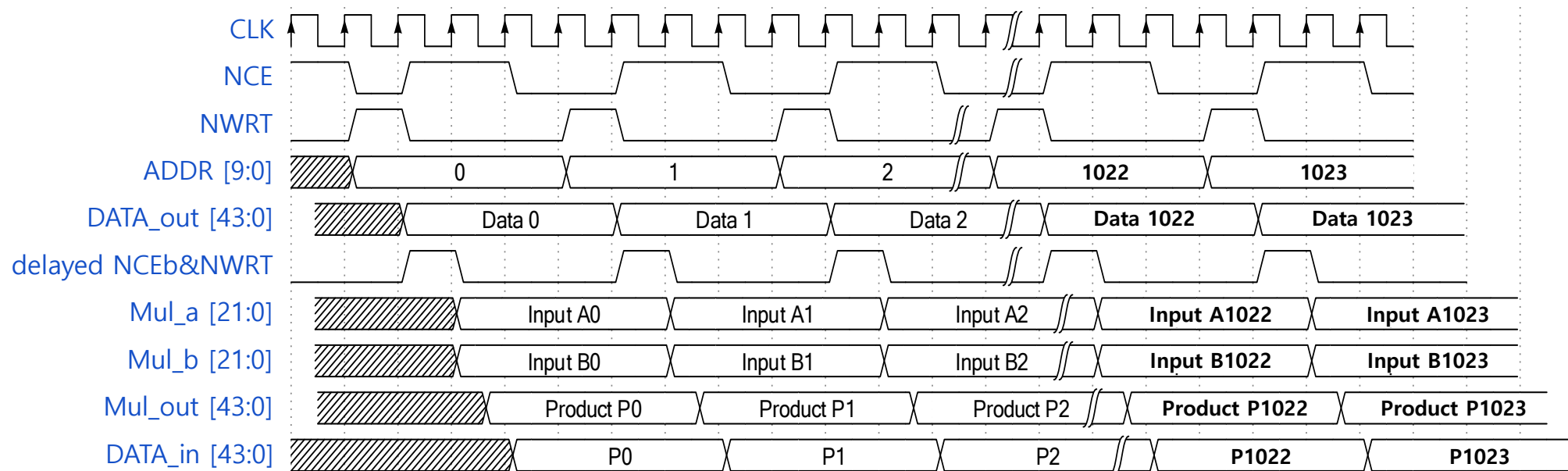


Memory control with 22×22 multiplier



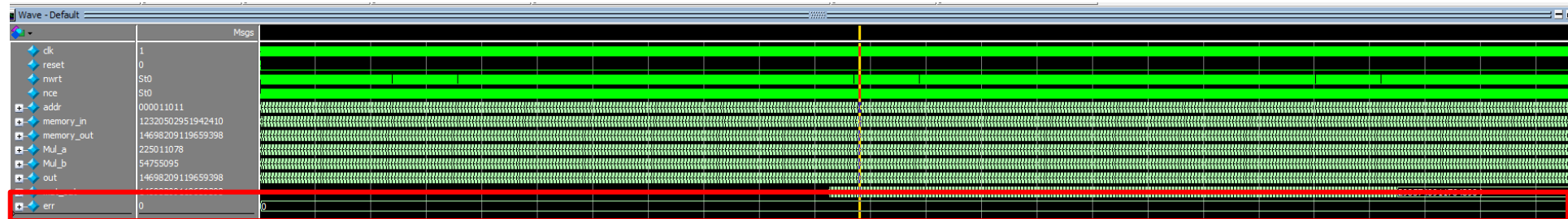
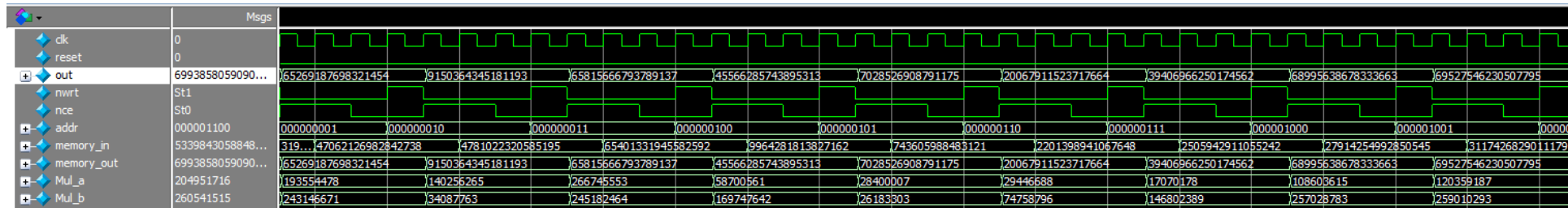
Memory control with 22×22 multiplier

Timing Diagram



- **Simulation results**

- Modify the previous stimulus files (input output files are given.)**



- Check the value of error counting number equals 'zero'**

References

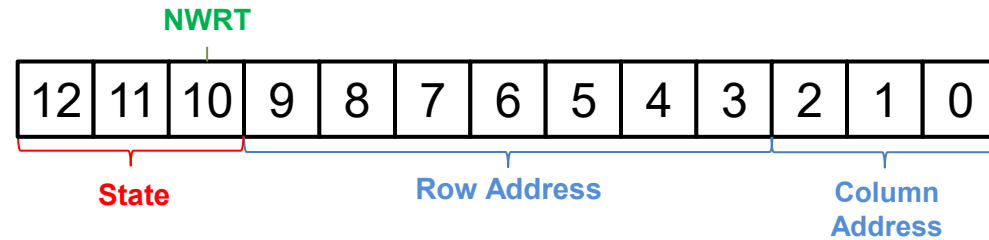
1) Bin, Hex, Dec

44'b0000_0000_0000_....
44'b1111_1111_1111_....
44'b0101_0101_0101_....
44'b1010_1010_1010_....

44'h000_0000_0000
44'hFFF_FFFF_FFFF
44'h555_5555_5555
44'hAAA_AAAA_AAAA

44'd000000000000000
44'd17592186044415
44'd5864062014805
44'd11728124029610

2) Counter



3) Instantiation

```
top_mbist MBIST_TEST(.MBIST_done(out_done), .DATA_out(out), .MBIST_start(MBIST_start), .clk(clk), .rstn(rstn));
```

```
module TOP_MBIST(MBIST_done, DATA_out, MBIST_start, clk, rstn);
```

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Check results and Submit report

- Check the synthesis result
 - No need to Synthesize !!
- Submit result report
 - Show the timing diagram
 - Results analysis
- Due : Before the next class (04/22 2:59PM).
Submit to **the TA in charge of your group.**